Collaborative Drawing

A collaborative drawing canvas for classrooms, built on the LiaScript layer API. Everyone draws on an own layer. Without a classroom you only see your own drawing, it is stored locally and is still there after a reload. As soon as you join a classroom, your layer is synchronized with all others.

Try it on LiaScript:

https://liascript.github.io/course/?
https://raw.githubusercontent.com/LiaTemplates/CollaborativeDrawing/main/README.md

Load the macros via import and set the persistence flag to true, so the drawings keep being updated while you are on another slide.

```
import: https://raw.githubusercontent.com/LiaTemplates/CollaborativeDrawing
  /README.md

persistent: true
```

This version needs LiaScript 2.5.0 or newer (`LIA.classroom.layer`). The first version, which only synchronizes drawings between connected participants (pubsub, nothing is stored), stays available as tag `0.0.1`:

```
import: https://raw.githubusercontent.com/LiaTemplates/CollaborativeDrawing
  /README.md
```

`@Collaborative.lines`

Draw lines with the mouse, a pen or a finger. Width and height define the resolution, the canvas always uses the full available width.

`@Collaborative.lines(width,height)`



With background image

The optional third parameter is a background image. The link style is preferred, since relative paths get resolved then.

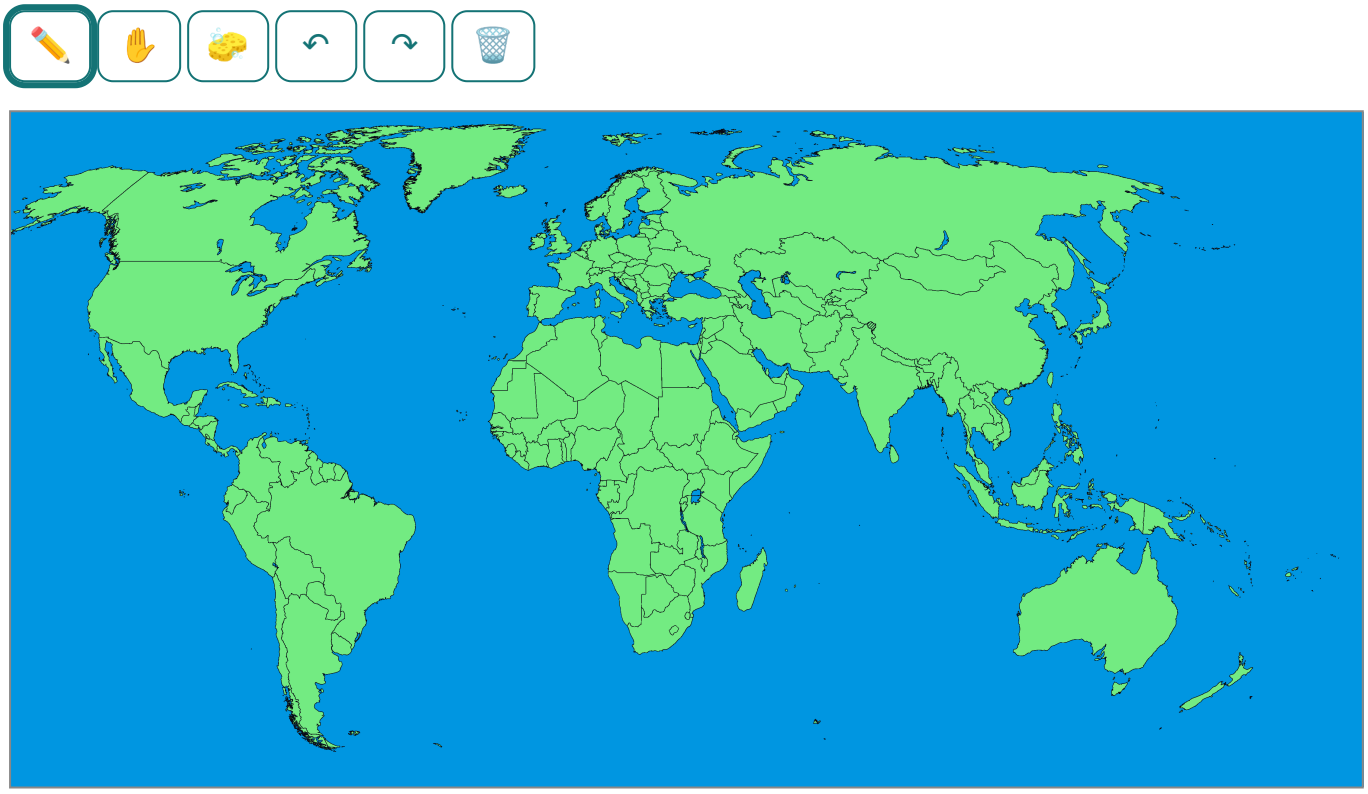
```
@[Collaborative.lines(640,320)](url)
```



`@Collaborative.dots`

Every click or tap places a dot. Combined with a background image this can be used to mark places on a map.

`@[Collaborative.dots(width,height)](url)`



Tools

Above every canvas there is a small toolbar. The tools change your own layer, the drawings of others stay untouched. Only the owner of a classroom can also move and erase the strokes of all others, and they see these changes on their own canvas.

Tool	What it does
✏️	draw lines, or place dots in `@Collaborative.dots`
✋	drag one of your strokes to another place
🧽	click one of your strokes to remove it
↶ ↷	undo and redo your last steps, also `Ctrl+Z` and `Ctrl+Y` on the canvas
🗑️	clear your own drawing, this can be undone as well

Undo works per stroke: while you draw, the others see the line growing, when you release it becomes one stroke, which is undone as a whole. The undo history lasts until the page is reloaded, or until the owner changes your drawing, so that an undo cannot take back the owner's change.

Who sees what?

Visibility follows the classroom mode, in the same way as quiz answers.

Situation	You see
no classroom	your own layer
classroom, Shared	all layers
classroom, Summary	your own layer, the owner sees all layers
classroom, Details	your own layer, the owner sees all with names
after leaving	your own layer, including what you drew in the classroom

Every person draws in the same color and with the same emoji as in the chat, the buttons below a canvas name who drew what. A click on a name hides or shows that drawing, `all` and `none` switch all of them at once. This only changes your own view, for example to look at the solutions of a class one by one, and hidden drawings cannot be moved or erased.

Implementation

The whole logic lives in the `@onload` block of this file. It only uses the layer part of the LiaScript classroom API.

```
// an own layer per peer, stored locally and synchronized in a classroom,
// named "<template>:<name>", so it does not mix with other templates
const layer = LIA.classroom.layer('CollaborativeDrawing:' + id)

// called with all visible layers, whenever one of them changes
layer.on((layers) => {
  // layers = [{ id, name, me, editable, color, symbol, values: [...] }]
})

// one value per stroke: [x, y] for a dot, [x1, y1, x2, y2, ...] for a line
layer.push(stroke)

// undo, erase, move: read the own layer, change it, write it back
const own = layer.get()
layer.set(own.slice(0, -1))

// the classroom owner may also change the layers of others,
// layers[i].editable tells which ones
layer.set(values, layers[i].id)
```

`set` only sends what actually changed: removing the last stroke is a single delete for the others, not a copy of the whole drawing.

Layer names and pubsub topics are shared by all templates of a course. That is why the name starts with the template, `CollaborativeDrawing:`, and on purpose the same name lets several templates work on the same drawing. The rest of the name is built from `@uid`, so it stays the same for everyone using the same version of a course. If you reorder canvases in a course, previously drawn layers are no longer found.